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The central pattern generator underlying swimming in *Dendronotus iris*: a simple half-center network oscillator with a twist

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Submitted 18 February 2016; accepted in final form 14 July 2016

Sakurai A, Katz PS. The central pattern generator underlying swimming in *Dendronotus iris*: a simple half-center network oscillator with a twist. *J Neurophysiol* 116: 1728–1742, 2016. First published July 20, 2016; doi:10.1152/jn.00150.2016.—The nudibranch mollusc, *Dendronotus iris*, swims by rhythmically flexing its body from left to right. We identified a bilaterally represented interneuron, Si3, that provides strong excitatory drive to the previously identified Si2, forming a half-center oscillator, which functions as the central pattern generator (CPG) underlying swimming. As with Si2, Si3 inhibited its contralateral counterpart and exhibited rhythmic bursts in left-right alternation during the swim motor pattern. Si3 burst almost synchronously with the contralateral Si2 and was coactive with the efferent impulse activity in the contralateral body wall nerve. Perturbation of bursting in either Si3 or Si2 by current injection halted or phase-shifted the swim motor pattern, suggesting that they are both critical CPG members. Neither Si2 nor Si3 exhibited endogenous bursting properties when activated alone; activation of all four neurons was necessary to initiate and maintain the swim motor pattern. Si3 made a strong excitatory synapse onto the contralateral Si2 to which it is also electrically coupled. When Si3 was firing tonically but not exhibiting bursting, artificial enhancement of the Si3-to-Si2 synapse using dynamic clamp caused all four neurons to burst. In contrast, negation of the Si3-to-Si2 synapse by dynamic clamp blocked ongoing swim motor patterns. Together, these results suggest that the *Dendronotus* swim CPG is organized as a “twisted” half-center oscillator in which each “half” is composed of two excitatory-coupled neurons from both sides of the brain, each of which inhibits its contralateral counterpart. Consisting of only four neurons, this is perhaps the simplest known network oscillator for locomotion.

virtual pharmacology; gastropod; locomotion; neural circuit; invertebrate; identified neuron

NEW & NOTEWORTHY

We have shown that the central pattern generator (CPG) underlying rhythmic swimming of a nudibranch mollusc is a half-center oscillator consisting of only two pairs of neurons that inhibit their contralateral counterparts. The two neurons in each half of the oscillator are contralateral to each other, giving the CPG a “twisted” organization. “Virtual pharmacology” experiments, using dynamic clamp, demonstrated that excitatory synapses within each half of the oscillator are necessary for producing rhythmic bursting.

RHYTHMICALLY ALTERNATING MOVEMENTS of body parts or whole body flexions are widely used in locomotor behaviors throughout the animal kingdom. To account for the production of rhythmic activity, Brown (1911) first proposed the concept of a half-center oscillator, in which bilaterally arranged neuronal components form mutually inhibitory connections to generate alternating activity. In the central pattern generator (CPG) for vertebrate locomotion, the half-center configuration plays a major role in producing rhythmic left-right and flexor-extensor alternation (Grillner and El Manira 2015; Kiehn 2016; Moulton et al. 2013). In addition to the mutually inhibitory connections, excitatory components in each half of the bilateral neural network also are crucial for providing drive and synchronicity (Buchanan and Grillner 1987; Kiehn et al. 2008; Li and Moulton 2012; Li et al. 2007, 2009). However, because of the numerical complexity of vertebrate CPGs, it has been notoriously difficult to delineate their functional composition (Marder et al. 2014). In that regard, molluscs provide excellent preparations for studying neuronal mechanisms underlying rhythmic behavior because of the small number of neurons comprising their CPGs (Arshavsky et al. 1997, 1998). We provide an example of a half-center oscillator in the nudibranch, *Dendronotus iris*, that is formed by only four interneurons, which are not endogenous bursters. We demonstrate that the excitatory synaptic drive within each half plays a crucial role in producing rhythmic activity in left-right alternation.

Dendronotus iris exhibits swimming behavior that consists of a series of laterally directed body flexions in which the head and tail alternately come together on the left and right sides (Sakurai et al. 2011). In a previous study, we proposed that the *Dendronotus* CPG might be one of the simplest CPGs, consisting only of a pair of mutually inhibitory neurons, named swim interneuron 2 (Si2). In this study, we found an additional pair of the neurons, which we named swim interneuron 3 (Si3). We show that these four neurons are necessary and sufficient for producing the rhythmic motor pattern. Moreover, Si3 forms a strong excitatory synaptic onto the contralateral Si2, which plays a crucial role in producing the rhythmic bursting activity by providing strong excitatory coupling between the two pairs of mutually inhibitory neurons to form the half-center oscillator.

MATERIALS AND METHODS

Animal collection, maintenance, and dissection. Specimens of *D. iris* (Mollusca, Gastropoda, Nudibranchia), 6–20 cm in body length,

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were collected as adults by Monterey Abalone (Monterey, CA) and Living Elements (Delta, BC, Canada). Animals were kept in artificial seawater tanks at 10–12°C on a 12:12-h light-dark cycle.

To isolate the brain, animals were anesthetized by injection of 0.33 M MgCl₂ into the body cavity. An incision was made in the body wall near the esophagus. The brain, consisting of the cerebral, pleural, and

pedal ganglia (Fig. 1A), was removed by cutting all nerve roots. The brain was transferred to a Sylgard-lined dish, where it was superfused with normal saline (in mM: 420 NaCl, 10 KCl, 10 CaCl₂, 50 MgCl₂, 11 D-glucose, and 10 HEPES, pH 7.6) or with artificial seawater (Instant Ocean, Mentor, OH) at a rate of 0.5 ml/min. The brain was pinned to the dish with the dorsal side up and with the pedal ganglia and pedal commissures positioned anterior to the cerebral ganglia. Connective tissue surrounding the brain was manually removed with forceps and fine scissors while the brain was kept at 4°C to reduce neuronal activity. The temperature was raised to 10°C before electrophysiological experiments were started.

Electrophysiology. Intracellular recordings were obtained using glass microelectrodes (15–35 MΩ) filled with 3 M KCl. Up to six microelectrodes were used simultaneously. Each electrode was connected to an Axoclamp 2B amplifier (Molecular Devices, Sunnyvale, CA). The membrane potential waveforms were digitized (>2 kHz) with a 1401Plus or Micro1401 analog-to-digital converter from Cambridge Electronic Design (Cambridge, UK). Suction electrodes made from polyethylene tubing were placed on both the left and right third pedal nerves (PdN3) for extracellular recording of nerve impulses and stimulation.

To evoke the swim motor program, either the left or right PdN3 was stimulated with a train of voltage pulses (5–15 V, 1 ms) at 5 Hz for 1–3 s via a suction electrode. The effect on burst period of current injection (from –5 to 5 nA) into the swim interneurons was examined using a bridge-balanced microelectrode. To test for monosynaptic connectivity and electrical coupling between the swim interneurons, high-divalent cation (Hi-Di) saline was used to raise the threshold for spiking and reduce spontaneous neural firing. The composition of the Hi-Di saline was (in mM) 285 NaCl, 10 KCl, 25 CaCl₂, 125 MgCl₂, 11 D-glucose, and 10 HEPES, pH 7.6. In some experiments, calcium-free Hi-Di saline was used to block chemical synaptic transmission. In Ca²⁺-free saline, CaCl₂ was replaced by MgCl₂. To measure the strengths of electrical connections, the membrane potential was manipulated by injecting a steady current through an additional microelectrode placed in the same neuron or under discontinuous current-clamp mode. To measure the time course of excitatory postsynaptic current evoked by Si3 in Si2, two-electrode voltage clamp was performed in Hi-Di saline.

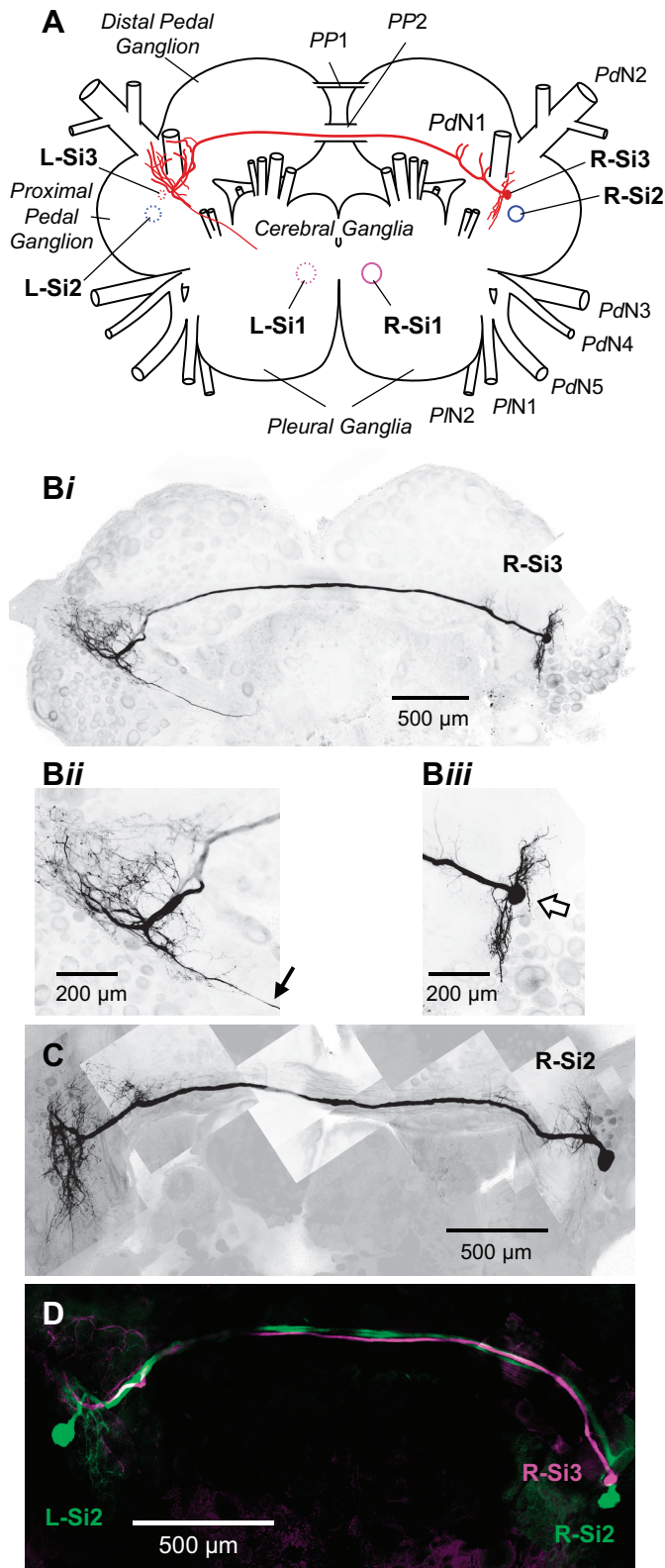


Fig. 1. Neuroanatomy of newly identified swim interneuron 3 (Si3) compared with that of previously identified Si1 and Si2. **A**: schematic drawing of the brain of *Dendronotus iris* showing the location of the somata of 3 bilateral pairs of the swim interneurons (L-/R-Si1, L-/R-Si2, and L-/R-Si3). The axonal projection of the right Si3 (R-Si3) is drawn in red. The *Dendronotus* brain consists of bilaterally arranged ganglionic lobes: the cerebral ganglia, the pleural ganglia, and pedal ganglia. The cerebral and pleural ganglia are fused to form the cerebropleural ganglia. Each pedal ganglion has two lobes: a proximal lobe, which is next to the cerebral ganglion, and a distal lobe from which the pedal commissures, PP1 and PP2, emerge. Nerves emerging from the pedal (Pd) and pleural (PI) ganglia are numbered from anterior to posterior and dorsal to ventral, such as PdN1–5 and PIN1 and 2. Nomenclature is based on Newcomb et al. (2006). Si1 has its soma in the cerebropleural ganglion, whereas Si2 and Si3 have their somata in the same vicinity in the proximal pedal ganglion. They each project an axon to the contralateral pedal ganglion through PP2 (see below). **B**: a photomontage of negative fluorescent images of R-Si3. It projects an axon to the contralateral pedal ganglion (i). Extensive branching can be seen in the terminal in the left pedal ganglion (ii). A small branch projects to the left cerebral ganglion (iii, arrow). There is a dendritic arbor near the soma (iii, open arrow). **C**: a photomontage of negative fluorescent images of R-Si2 labeled by biocytin injection. The soma is located on the dorsal surface of the proximal pedal ganglion. It also projects an axon to the contralateral pedal ganglion and forms an extensive arborization (Sakurai et al. 2011). **D**: a photomontage showing L-Si2, R-Si2, and R-Si3. All 3 neurons have axons that run side-by-side. L- and R-Si2 (green) were labeled by intracellular injection of Lucifer yellow. R-Si3 (magenta) was labeled by biocytin reacted with streptavidin-Alexa Fluor 594 conjugate.

Data acquisition and analysis were performed with Spike2 software (CED, Cambridge, UK), SigmaPlot (Jandel Scientific, San Rafael, CA), and MATLAB (The MathWorks, Natick, MA).

Tracer injections and immunohistochemistry. To label neurons with biocytin or Lucifer yellow (Sigma, St. Louis, MO), a microelectrode filled with either 2% biocytin in a 1 M KCl solution or 2% Lucifer yellow in a 1 M LiCl solution was inserted into the soma, and current pulses (-5 nA, 1 Hz, 50% duty cycle) were applied for >1 h. For Lucifer yellow injection, pressure pulses (1 Hz, duration 5–15 ms at 50 psi) were applied to the pipette through a Picospritzer III (Parker-Hannifin, Fairfield, NJ) together with the current pulses. The preparation was then superfused in physiological saline for 6 h at 10°C . The brain was fixed in 4% paraformaldehyde in 0.1 M phosphate-buffered saline (PBS; pH 7.4) overnight at 4°C . After being rinsed with PBS several times, the brain was treated with 4.0% Triton-X 100 (Sigma) in PBS for 2–4 h and then incubated for 2–5 days at 4°C in streptavidin-Alexa Fluor 594 conjugate (Invitrogen, Carlsbad, CA) diluted 1:200 in PBS containing 0.5% Triton-X 100. The brain was washed 6 times with PBS over 6 h, dehydrated in a graded ethanol series, cleared by methyl salicylate, and mounted on a glass slide with Cytoseal 60 (Electron Microscopy Sciences, Washington, PA).

Fluorescence images were obtained using a confocal microscope (LSM700 on an Axio Examiner D1 microscope; Carl Zeiss, Thornwood, NY) with a $\times 10$ or $\times 20$ objective. Fluorophores were excited with one of two lasers (488 or 543 nm), and fluorescent emissions were bandpass filtered (505–550 nm) for visualization of Lucifer yellow and long-pass filtered (560 nm) to visualize Alexa 594. Maximal projections of confocal stacks were exported as TIFF files and imported into Adobe Photoshop CS (Adobe, San Jose, CA). In Photoshop, projections were assembled into a montage of the entire central nervous system, and overall brightness and contrast were adjusted.

Cross-correlation. To determine the correlation of the impulse activity in a nerve root to the neuronal spiking, we performed a cross-correlation analysis using Spike2 software (CED). A cross-correlation counts the number of nerve impulses in every time bin after each neuronal spike. Each neuronal spike generates one sweep of analysis. The zero times of each sweep are aligned, and the counts are accumulated to form a histogram. The value is then normalized to the maximal value in each preparation and averaged across preparations (see Fig. 3, C and D).

To detect bursts in a spike train, a cross-correlogram was made from the spike trains in the right Si2 against spikes in the left Si2. The bin size was set to 10% of the cycle period in the nerve-evoked swim motor pattern. When the Si2 pair exhibited bursts in left-right alternation, the cross-correlogram showed a deepest trough at the first bin, which was followed by a peak at bins near 50% of the burst cycle period. When the peak value was two times larger than the trough value, the spiking activity was considered as bursting. A train of Si2 spikes was then counted as a burst if the spike interval was <0.5 s and the duration of the spike train was >0.5 s. When Si2s were suppressed from spiking, the Si3s were used to measure the burst frequency (cf. Fig. 7C). To analyze burst-phase relationships, the start, end, and median spike times for each burst were calculated from spike times using Spike2 script (CED). Phases were defined with respect to the median spike in the right Si2. There was no consistent asymmetry in bursting; therefore, if there was no recording from the right Si2, the left and right were reversed for all of the traces, and phase was defined with respect to the left Si2.

Dynamic clamp. To perform dynamic clamping, each Si3 was impaled with a single electrode to measure membrane potential, and each Si2 was impaled with two microelectrodes, one for membrane potential recording and the other for current injection (see Fig. 10A). Dynamic clamp software StdC (Kemenes et al. 2011) was used to enhance or suppress Si3-evoked excitatory postsynaptic potentials (EPSPs) in Si2. The current injected into the postsynaptic Si2, I_{stim} ,

was calculated in each dynamic clamp cycle using a first-order kinetics model of the release of neurotransmitter (Destexhe et al. 1994; Kemenes et al. 2011; Sharp et al. 1996):

$$I_{\text{syn}} = g_{\text{syn}} S(t) [V_{\text{syn}} - V_{\text{post}}(t)], \quad (1)$$

where $S(t)$ is the instantaneous synaptic activation, g_{syn} is the maximum synaptic conductance, and V_{syn} is the reversal potential (0 mV) of the synapse. To counteract Si3-evoked synaptic potentials, a negative value was used for g_{syn} .

The instantaneous activation $S(t)$ is given by the differential equation

$$[1 - S_{\infty}(V_{\text{pre}})] \tau_{\text{syn}} \frac{ds}{dt} = [S_{\infty}(V_{\text{pre}}) - S(t)], \quad (2)$$

where

$$S_{\infty}(V_{\text{pre}}) = \begin{cases} \tanh \left[\frac{V_{\text{pre}} - V_{\text{thresh}}}{V_{\text{slope}}} \right] & \text{if } V_{\text{pre}} > V_{\text{Thresh}} \\ 0 & \text{otherwise} \end{cases}. \quad (3)$$

S_{∞} is the steady-state synaptic activation and τ_{syn} is the time constant for synaptic decay. The Si3-evoked excitatory postsynaptic current recorded in Si2 under voltage clamp was 18.6 ± 1.5 ms (mean $\pm$ SD, $n = 3$). However, in this study τ_{syn} was set to 40 ms to make the injected current more effective without injecting large-amplitude currents that often became harmful and deteriorated recording conditions. V_{pre} is the presynaptic membrane potential of Si3, and V_{thresh} is the threshold potential for the release of neurotransmitter, which was set to the level of 50% height of the smallest Si3 action potentials. The synaptic slope parameter of the activation curve (V_{slope}) was set to 25 mV. g_{syn} was varied between -240 and 120 nS.

Statistics. Statistical comparisons were performed using SigmaPlot (ver. 12.5; Jandel Scientific, San Rafael, CA) for Student's t -test, paired t -test, and one-way ANOVA on ranks with Dunn's method. The Shapiro-Wilk test was used to assume normality of data structure. In all cases, $P < 0.05$ was considered significant. For t -tests, a two-tailed P value was used. Results are means $\pm$ SD.

RESULTS

Identification of swim interneuron 3. In this study, we identified a bilaterally represented neuron that is a member of the swim CPG in *D. iris*. We named this neuron swim interneuron 3 (Si3). There was a single Si3 soma located in the proximal lobe of each pedal ganglion, one cell layer below the dorsal surface, near pedal nerve 1 (PdN1; Fig. 1A). Intracellular labeling with biocytin revealed that the Si3 soma was $36\text{--}74$ μm in diameter (55.2 ± 10.3 μm , $n = 9$). As with Si2, Si3 had a primary axonal process that projected to the contralateral pedal ganglion through the large pedal commissure (PP2; Fig. 1, A and Bi). In all preparations ($n = 9$), the Si3 axon terminal reached to the proximal lobe of the pedal ganglion contralateral to its soma. It had dense branching both at its axon terminal in the contralateral pedal ganglion (Fig. 1Bii) and near its soma, which formed a characteristic t-shape (Fig. 1Biii). In 7 of 9 preparations, one of the Si3 axonal neurites was observed to extend to the cerebral ganglion (arrow in Fig. 1Bii). After an extensive intracellular survey of the dorsal surface of the pedal ganglia, we could not find any other neurons that were rhythmic members of the swim CPG besides Si2 and Si3.

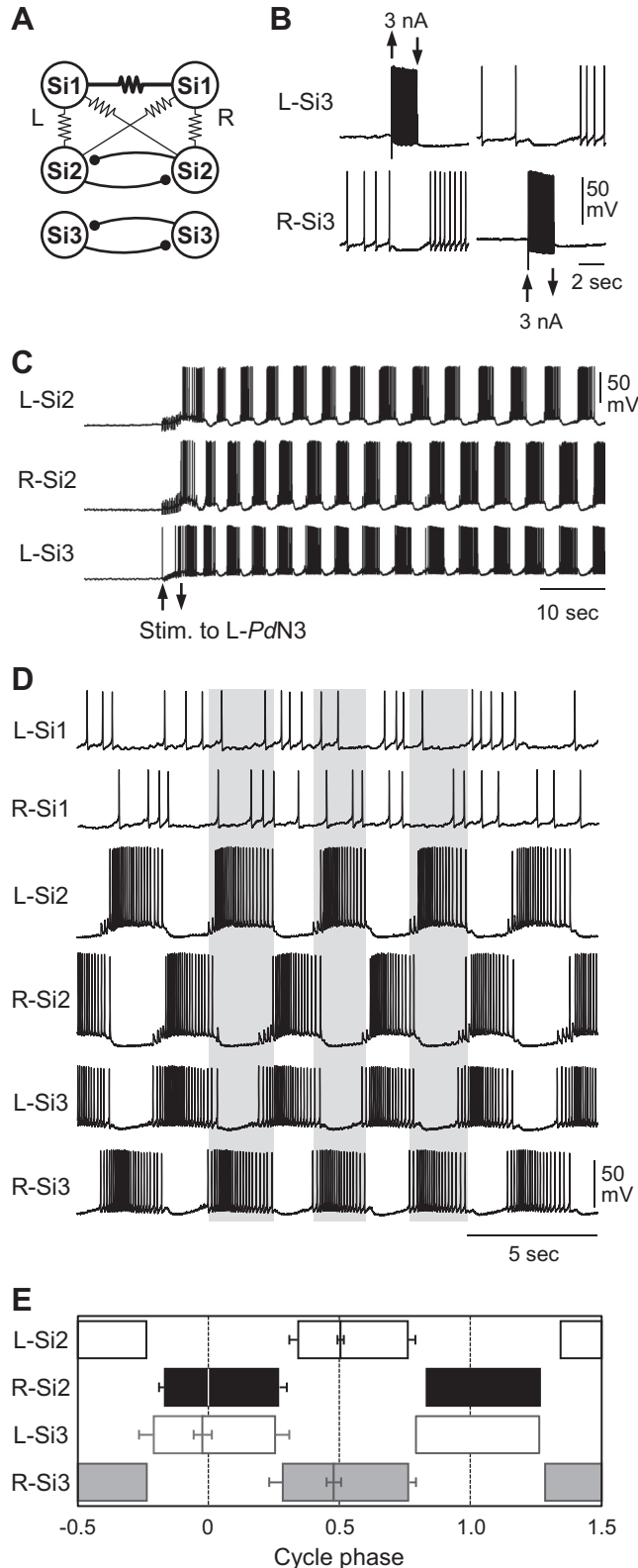
Si2 and Si3 had their somata near each other, and both had axons that project through PP2 (Fig. 1, A, Bi, and C). Labeling of Si2 and Si3 with different markers revealed that the main axons of Si2 and Si3 ran adjacent to each other throughout PP2

and the distal pedal ganglia (Fig. 1D; $n = 3$). Nevertheless, their neuroanatomy had several distinguishing features. The soma of Si2, which ranged in diameter from 60 to 102 μm ($81 \pm 12.6 \mu\text{m}$, $n = 13$), was significantly larger than the Si3 soma ($P < 0.05$ by Student's t -test). Unlike the Si3 soma, the

Table 1. Phases and duty cycles of Si2s and Si3s

Neuron	Phase	Duty Cycle	Intraburst Spikes, spikes/s	n
L-Si2	0.51 ± 0.01	0.44 ± 0.05	7.1 ± 2.1	29
R-Si2	0	0.45 ± 0.03	7.0 ± 2.0	27
L-Si3	-0.02 ± 0.04	0.48 ± 0.05	6.7 ± 2.0	18
R-Si3	0.48 ± 0.03	0.50 ± 0.06	7.1 ± 2.0	20

Values are means $\pm$ SD in left (L) and right (R) swim interneurons 2 (Si2) and 3 (Si3); n = no. of preparations.



Si2 soma had no fine branches nearby and was located on the surface layer of cell bodies (Fig. 1C; Sakurai et al. 2011).

Si3 is rhythmically active in a constant phase relation to Si2. It was shown previously that the Si2 pair exhibit reciprocal inhibition and fire bursts in alternation, whereas the Si1 pair have only electrical connections (Fig. 2A) and fire irregularly throughout the swim motor pattern (Fig. 2D; Sakurai et al. 2011). Similar to the Si2 pair, the Si3 pair exhibited reciprocal inhibition (Fig. 2, A and B). A train of spikes in one Si3 hyperpolarized the other Si3, inhibiting its spiking activity. Furthermore, like Si2, Si3 was rhythmically active, firing bursts of action potentials when the swim motor pattern was triggered by stimulating a body wall nerve (Fig. 2C).

Throughout the swim motor pattern, the Si3 bursts were time-locked to the contralateral Si2 bursts. The Si3 bursts led the contralateral Si2 bursts by a few spikes (Fig. 2D) or were coincident with them (see Fig. 3B). The phases and duty cycles of Si2 and Si3 relative to the median spike of the right Si2 are diagrammed in Fig. 2E. The periods of the swim motor patterns in these recordings ranged from 3.4 to 8.6 s with the average period being 5.2 ± 1.3 s (12.3 ± 2.9 bursts/min, $n = 31$). Within each burst, Si2 fired spikes at 3.7–11.7 Hz with an average of 7.0 ± 2.0 Hz ($n = 56$), whereas Si3 fired at 3.5–11.3 Hz, averaging 6.9 ± 2.0 Hz ($n = 38$). Si2 and Si3 each fired bursts $\sim 50\%$ out of phase from their own contralateral counterpart. The duty cycle of Si3 tended to be larger than that of Si2, sometimes exceeding 50%, meaning that it overlapped in firing with its contralateral counterpart (Fig. 2D). Data are summarized in Table 1.

The bursts of Si2 and Si3 are oppositely correlated to body wall nerve activity. Despite having similar axonal projections and soma locations, the ipsilateral Si2 and Si3 burst in opposite phases (Fig. 2, C–E). To determine which side of the efferent nerve outputs corresponds to Si2 and/or Si3 activities, we

Fig. 2. Si3 inhibits its contralateral counterpart and bursts in phase with the swim motor pattern. A: the previously published connectivity of swim interneuron 1 (Si1) and swim interneuron 2 (Si2). Lines terminating in filled circles indicate inhibitory synapses. Resistor symbols indicate electrical connections. The thickness of the line indicates the strength of connection. B: the Si3 pair forms mutually inhibitory synapses. Spiking in one Si3 produced a sustained hyperpolarization in its contralateral counterpart. C: rhythmic bursting in Si2 and Si3 was induced by extracellular stimulation (5 Hz for 3 s) of body wall nerve (left PdN3) as indicated by arrows. D: simultaneous intracellular recordings from all 6 interneurons during the swim motor pattern showed that Si2 and Si3 produced alternating bursts of activity on the left and right sides. Gray bars indicate the durations of 3 consecutive R-Si3 bursts. E: a phase diagram of the bursts for L-Si2, R-Si2, L-Si3, and R-Si3 shows that the contralateral Si2 and Si3 fired concurrently. The median spike of R-Si2 was set to be the phase zero. Bars run from the average phase for the first spike to the average phase of the last spike for each burst. The central vertical bar is the averaged phase of the median spike in each burst. The error bars are standard deviation. Sample sizes are listed in Table 1.

recorded from them while simultaneously recording multiunit activity from each PdN3, which projects unilaterally to the body wall musculature (Fig. 3). During the swim motor pattern, the impulse activity in the pedal nerves waxed and waned

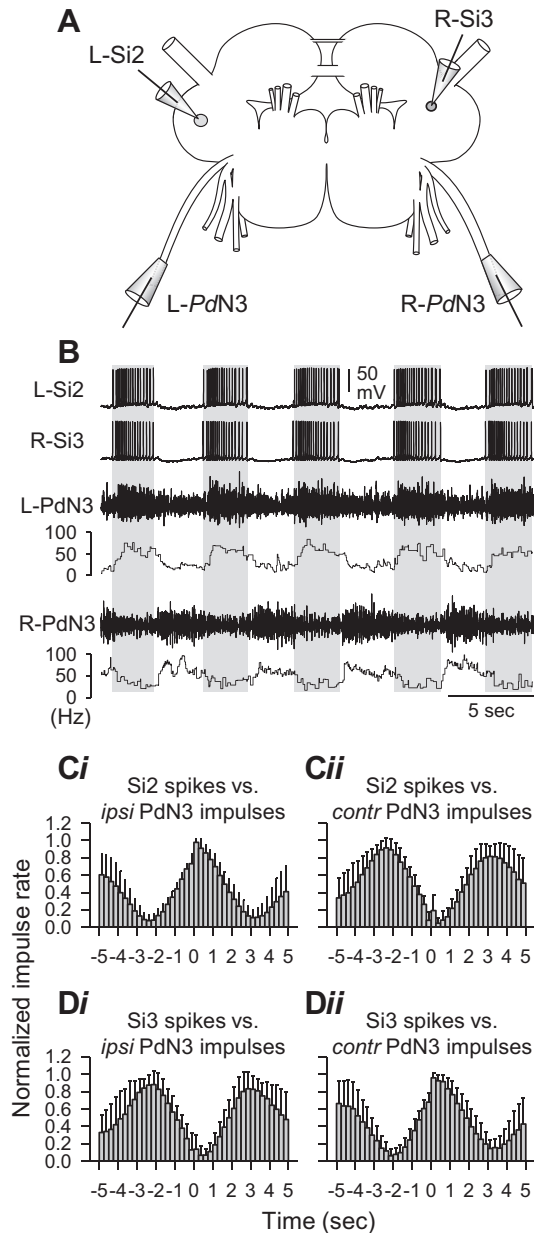


Fig. 3. Bursting activity in the interneurons corresponds to efferent activity ipsilateral to Si2. **A**: a schematic drawing of the *Dendronotus* brain shows the recording sites. **B**: simultaneous intracellular recordings were made from L-Si2 and R-Si3, and extracellular impulse recordings from L- and R-PdN3. In graphs below each nerve impulse recording, running averages of instantaneous impulse frequency (Hz) in 0.5-s windows are plotted. The amplitude and frequency of nerve impulses waxed and waned in left-right alternation. The synchronous bursts of L-Si2 and R-Si3 corresponded to the increase in the instantaneous impulse frequency recorded in L-PdN3. Cross-correlograms were made for nerve impulses recorded from PdN3 with ipsilateral Si2 spikes (**Ci**; $n = 18$), contralateral Si2 spikes (**Cii**; $n = 18$), ipsilateral Si3 spikes (**Di**; $n = 8$), and contralateral Si3 spikes (**Dii**; $n = 7$). The impulse counts were normalized to the maximal rate (counts/s) recorded in each preparation. The impulse activity in PdN3 showed strong positive correlation with the ipsilateral Si2 and negative correlation with the contralateral Si2. In contrast, Si3 spikes were negatively correlated with the ipsilateral PdN3 impulses and positively with the contralateral PdN3 impulses. ipsi, Ipsilateral; contra, contralateral.

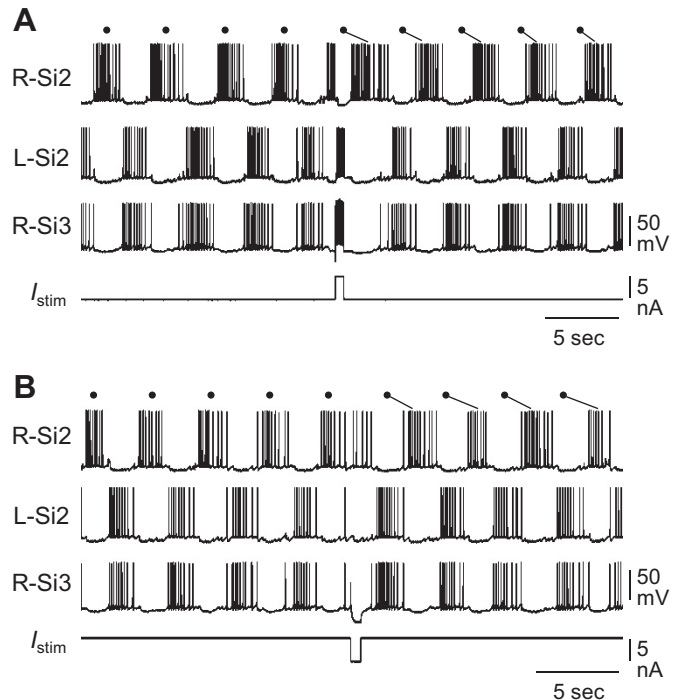


Fig. 4. Brief perturbation of Si3 resets the motor pattern. A brief depolarizing (5 nA, 1 s; **A**) or hyperpolarizing (−5 nA, 1 s; **B**) current pulse (I_{stim}) injected into R-Si3 phase-shifted the motor pattern. Dots above the traces represent the expected median spike times of the R-Si2 burst. In both cases, there was a persistent phase delay with no noticeable change in phase relationships after the initial perturbation. Traces in **A** and **B** were obtained from the same preparation.

in frequency over the period of a swim cycle (Fig. 3*B*; cf. Sakurai et al. 2011). Periodic increases in the impulse frequency recorded from PdN3 coincided with bursting in the ipsilateral Si2 and contralateral Si3 (Fig. 3*B*). The spike cross-correlogram revealed that Si2 spikes showed positive correlation with the ipsilateral nerve impulses and negative correlation with the contralateral nerve impulses (Fig. 3*C*). In contrast, the Si3 spikes showed a negative correlation with the ipsilateral nerve impulses and positive with the contralateral nerve impulses (Fig. 3*D*). Assuming that the efferent motor output produces muscle contractions, and hence body bending, these results suggest that during the swimming behavior, the animal bends in the direction ipsilateral to Si2 bursts and contralateral to Si3 bursts.

Both Si2 and Si3 are necessary components for the swim motor pattern. One hallmark of a CPG member is that manipulation of its activity perturbs the motor pattern. Injecting a brief depolarizing or hyperpolarizing current pulse into Si3 caused a phase shift in Si2 bursting in all preparations examined (Fig. 4; $n = 19$). In the example shown, a brief depolarization of one Si3 caused a phase delay by evoking a premature excitation of the contralateral Si2, which prevented the ipsilateral Si2 from firing (Fig. 4*A*). Once current injection in the Si3 ended, the ipsilateral Si2 started to burst again, well after when it would have fired. Injection of a hyperpolarizing current pulse into Si3 at a different phase also caused a delay in the subsequent bursts (Fig. 4*B*). The contralateral Si2 missed a burst while the ipsilateral Si2 continued to fire, extending its burst during until finally halted by the contralateral Si2. This

suggests that Si3 has strong connections with Si2 and hence plays an important role in motor pattern generation.

Prolonged injection of current revealed that both Si2 and Si3 play critical roles in motor pattern generation (Fig. 5). When just one Si2 was activated by depolarizing current injection, the swim motor pattern stopped in 60% of preparations ($n = 9$ of 15; Fig. 5A) or slowed down ($n = 6$ of 15). The extent of decrease in the Si2 burst frequency became larger when the other Si2 was forced to fire spikes at a higher rate (Fig. 5D). In contrast, when one Si2 was suppressed by a hyperpolarizing current injection, the bursting activity in the other three neurons persisted with a slight decrease in frequency in all preparations examined ($n = 13$; Figs. 5, B, D, and I). When spiking of both Si2s was suppressed by simultaneous injection of hyperpolarizing current, the Si3 pair alone did not continue bursting in 50% of the preparations ($n = 4$ of 8; Fig. 5C). In the other half of the preparations, the burst frequency was significantly reduced (Fig. 5I). Thus at least one of the two Si2s was necessary to produce rhythmic bursting together with the Si3 pair.

Unilateral manipulation of Si3 spiking had similar though more variable effects on the swim motor pattern than Si2 manipulation. As with Si2, intense tonic firing of one Si3 arrested the rhythmic activity in 77% of the preparations ($n = 10$ of 13; Fig. 5E). The decrease in the Si2 burst frequency became larger when one Si3 was forced to fire spikes at a higher rate (Fig. 5H). Suppression of spiking in just one Si3 halted the swim motor pattern in 72% of the preparations ($n = 13$ of 18; Fig. 5, F, H, and I). In the rest of the preparations, suppressing spiking in one Si3 had little effect on ongoing bursting in the other swim interneurons ($n = 5$ of 18). In some preparations, the effect was variable from time to time even within the same individual animal. The cause of such variation is currently unknown. Suppression of spiking in both Si3s halted the swim in all preparations examined ($n = 5$; Fig. 5, G and I).

Together, these results show that both Si2 and Si3 play critical roles in generating the swim motor pattern. Although the circuit can operate without one of the Si2 pair, both left and right Si3 were necessary in the majority of preparations to produce rhythmic bursting.

Activation of all four neurons is necessary and sufficient to induce a swimlike motor pattern. At rest, when there was no swim motor pattern, Si2 and Si3 exhibited low-frequency (<2 Hz), irregular spiking (Fig. 6). Constant-current injection induced tonic spiking, but there was no rhythmic periodicity in the firing frequency of either Si2 (Fig. 6A) or Si3 (Fig. 6B) in any of the preparations examined (Si2, $n = 22$; Si3, $n = 19$).

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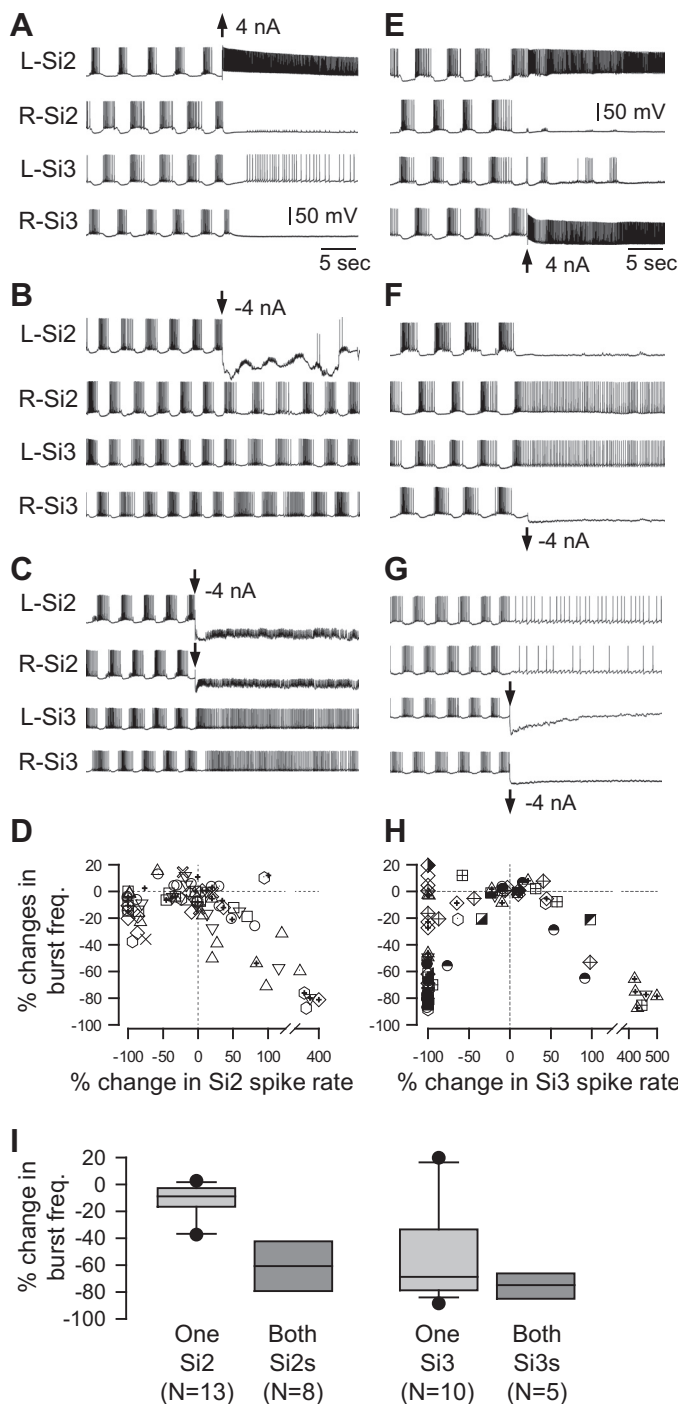


Fig. 5. Effects of prolonged current injection revealed that Si2 and Si3 are both necessary for generation of the rhythmic motor pattern. **A:** injection of 4-nA depolarizing current into L-Si2 caused it to fire tonically and immediately suppressed bursting in all neurons. **B:** in contrast, injection of -4-nA hyperpolarizing current caused little change in the rhythmic activity of other neurons. **C:** hyperpolarizing both the left and right Si2s caused the two Si3s to cease bursting and fire tonically. **D:** the relationship between the percent change in burst frequency and the percent change in spike rate of the Si2 that was manipulated by current injection. The burst frequency was measured from the contralateral Si2. Data were obtained from 22 preparations. Symbols indicate individual preparations. **E:** injection of 4-nA depolarizing current into R-Si3 halted or slowed bursting in all neurons. The Si2 contralateral to the stimulated Si3 fired tonically while the other Si2 became silent due to tonic inhibition from its counterpart. Even though R-Si3 was intensely firing, its counterpart L-Si3 fired bursts irregularly. **F:** injection of -4-nA hyperpolarizing current into R-Si3 halted the motor pattern with the contralateral Si2 silent and R-Si2 and L-Si3 firing tonically. **G:** hyperpolarizing both Si3s caused the Si2 pair to cease bursting and fire tonically. **H:** the relationship between the percent changes in burst frequency and the percent changes in spike rate of the Si3 that was manipulated by current injection shows a great deal of variability. The burst frequency was measured from the Si2 ipsilateral to the manipulated Si3. Data were obtained from 20 preparations. Symbols indicate individual preparations. **I:** box plot showing changes in burst frequency when spiking was suppressed in one Si2, both Si2s, one Si3, or both Si3s by injecting hyperpolarizing current (-4 nA). Suppression of one Si2 had significantly less effect on the swim cycle than the other manipulations ($P < 0.05$ by 1-way ANOVA on ranks with Dunn's method). freq., Frequency.

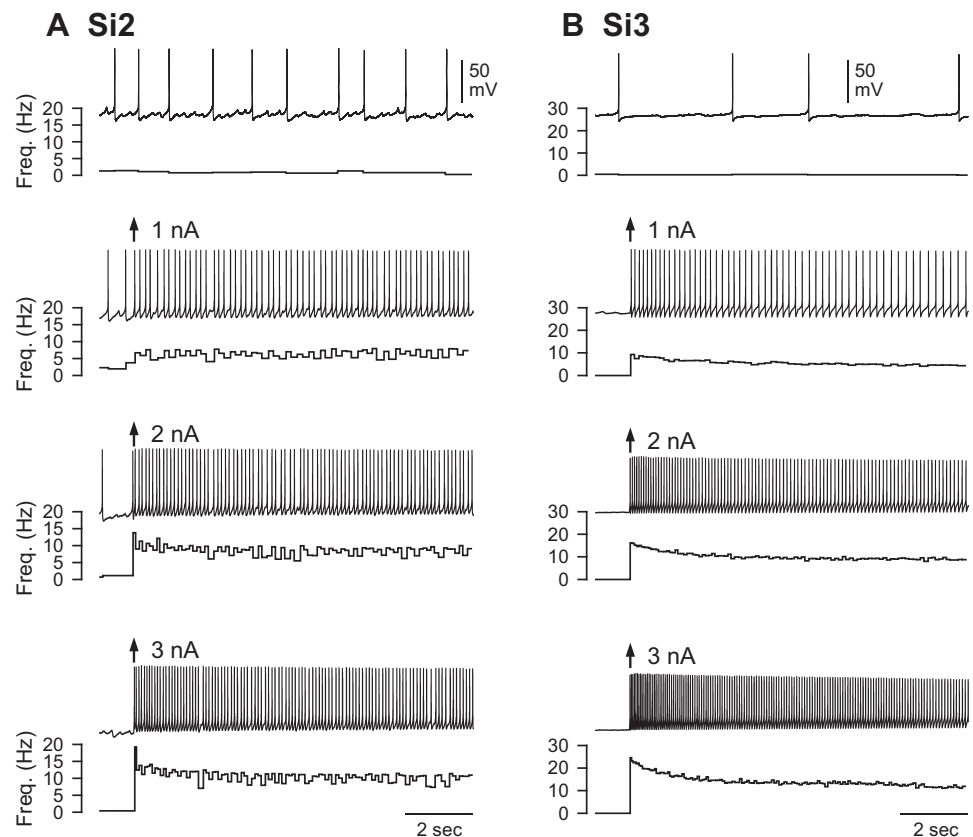


Fig. 6. Si2 and Si3 do not burst in response to depolarizing current steps. Constant-current steps were applied to Si2 (A) or Si3 (B) at varying intensities (0, 1, 2, and 3 nA). No oscillatory activity was induced by a steady current pulse. Both cells showed low levels of spontaneous spiking activity (*top* traces). When injected with constant current, Si3 fired at higher rate, had more spike frequency adaptation, and fired more regularly than Si2.

Bursting could be evoked by simultaneously depolarizing pairs of neurons (Fig. 7). Simultaneous depolarization of just the Si2 pair evoked bursting in left-right alternation in 90% of preparations ($n = 9$ of 10 preparations; Fig. 7A), which was limited to the Si2 pair; the Si3 pair exhibited mostly subthreshold membrane potential oscillations (Fig. 7Ai). The Si2 rhythmic activity was reflected in the impulses recorded in the pedal nerves (L- and R-PdN3), which waxed and waned in frequency over the period of the stimulation (Fig. 7Aii, *bottom* two traces). However, the burst frequency was significantly lower than the actual swim motor pattern triggered by nerve stimulation (Fig. 7Aiii). The maximal burst frequency of Si2 in each preparation evoked by current injection was 4.0 ± 2.7 bursts/min ($n = 12$), which was significantly lower than the average burst frequency of the swim motor pattern in the same preparation (10.7 ± 1.6 bursts/min, $n = 12$; $P < 0.001$ by paired t -test).

Simultaneously depolarizing just the Si3 pair caused rhythmic bursting in all four neurons that more closely resembled the swim motor pattern ($n = 8$ of 10 preparations; Fig. 7Bi). During current injection, the frequency of nerve impulses showed rhythmic waxing and waning reflecting the activity of the Si2 and Si3 pairs (Fig. 7Bii, *bottom* two traces). The burst frequency increased with increased amount of depolarizing current (Fig. 7Biii). The average maximal burst frequency in each preparation evoked by current injection was 9.1 ± 2.6 bursts/min ($n = 10$), which was slightly lower than the average burst frequency of the swim motor pattern in the same preparation (10.4 ± 1.5 bursts/min, $n = 10$), but there was no significant difference from it ($P = 0.12$ by paired t -test).

To test whether the Si3 pair alone can generate alternating activity, we next examined the effect of current injection into only the Si3 pair while suppressing Si2 spiking activity by injecting negative current (-2 nA; Fig. 7Ci). Both Si3 fired tonically during depolarizing current injection of up to 3 nA in 5 of 7 preparations (Fig. 7, Ci and Cii). In two preparations, they showed alternating slow bursting similar to that in the Si2s (cf. Fig. 7Ai). The average maximal burst frequency of Si3-evoked current injection was 1.7 ± 3.6 bursts/min ($n = 7$), which was significantly lower than that of the swim motor pattern in the same preparation (10.8 ± 1.0 bursts/min, $n = 7$; $P < 0.001$ by paired t -test).

When all four neurons were depolarized together, they exhibited rhythmic bursting that most closely resembled the swim motor pattern ($n = 9$; Fig. 7Di). During the current-evoked bursting, the frequency of impulses recorded from the body wall nerves (Fig. 7Dii, *bottom* two traces) was higher than when just one pair of neurons was depolarized (Fig. 7, Aii and Bii). The burst frequency increased with increased amount of depolarizing current (Fig. 7Diii). The average maximal burst frequency evoked by current injection current was 10.3 ± 1.1 bursts/min ($n = 7$), which was not significantly different from the average burst frequency of the swim motor pattern recorded from the same animal (10.1 ± 1.5 bursts/min, $n = 7$; $P = 0.46$ by paired t -test). Together, these results suggest that spiking in all four neurons is necessary and sufficient to produce the swim motor pattern.

Si3 drives the contralateral Si2. During the swim motor pattern, the Si3 burst often preceded the contralateral Si2 burst by a few spikes ($n = 18$ of 33 preparations), eliciting discrete synaptic potentials in the contralateral Si2 before it started to

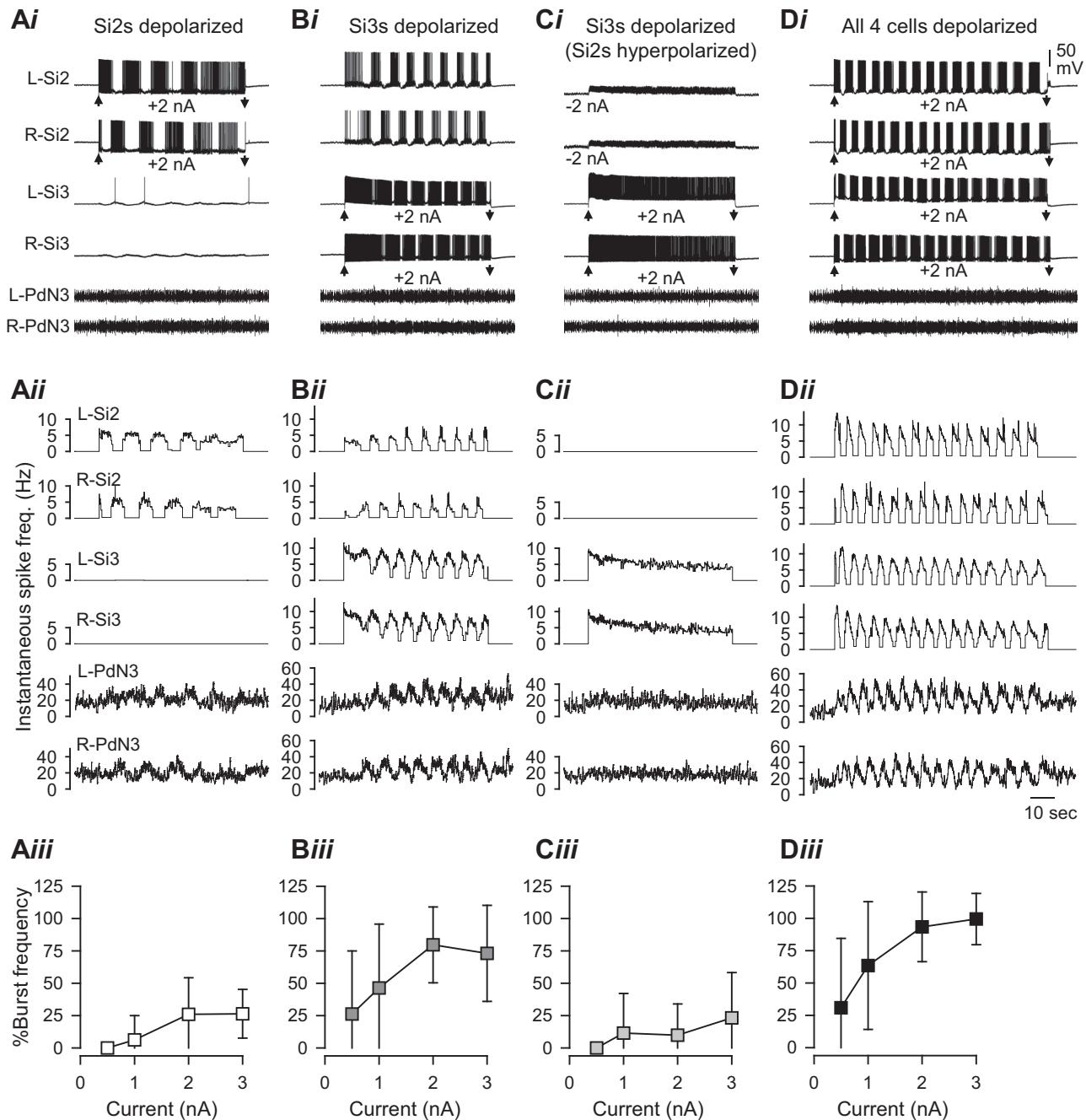


Fig. 7. Simultaneous depolarization of all Si2s and Si3s induces bursting activity that resembles the swim motor pattern. In *A–D*, simultaneous intracellular recordings were made from left and right Si2 and Si3 and extracellular recordings from left and right PdN3 (*i*). Instantaneous spike frequency is shown for each of the swim interneurons and nerves (*ii*). For nerve impulses, running averages of instantaneous impulse frequency (Hz) with a 0.5-s window are plotted. The relationship between the burst frequency during stimulation as a percentage of the swim cycle frequency and the amount of current injected into neurons is plotted (*iii*). Depolarization of both Si2s by injecting constant depolarizing current caused slow bursting in Si2, whereas Si3 showed a small membrane potential oscillation (*Ai* and *Aii*). In contrast, simultaneous depolarization of both Si3s caused all four neurons to burst (*Bi* and *Bii*). When Si2 was prevented from spiking by injection of a constant negative current, Si3 showed tonic spiking with no oscillation when depolarized (*Ci* and *Cii*). Injecting constant depolarizing current into all 4 neurons caused them to burst regularly (*Di* and *Dii*). Only when all 4 neurons were depolarized did the burst frequency and regularity reach that of the nerve-evoked swim motor pattern (*Diii*).

fire action potentials (Fig. 8*A*, see also Fig. 2*C*). Furthermore, hyperpolarization of Si2 by injection of a step current revealed a barrage of EPSPs that had a one-to-one correspondence to the spikes in the contralateral Si3 (Fig. 8*A*). Overlaying the traces showed that each Si3 spike evoked a large EPSP in the contralateral Si2 with a fixed latency while producing time-locked IPSPs in the contralateral Si3 (Fig. 8*B*; $n = 39$).

In addition to the excitatory synapse, there was strong electrical coupling between Si3 and the contralateral Si2 (Fig. 8*C*). The coupling coefficient from Si3 to the contralateral Si2 was 0.12 ± 0.05 ($n = 12$), and that from Si2 to the contralateral Si3 was 0.12 ± 0.06 ($n = 7$), indicating that there was no rectification in these electrical connections. In contrast, the coupling coefficient between ipsilateral Si2 and

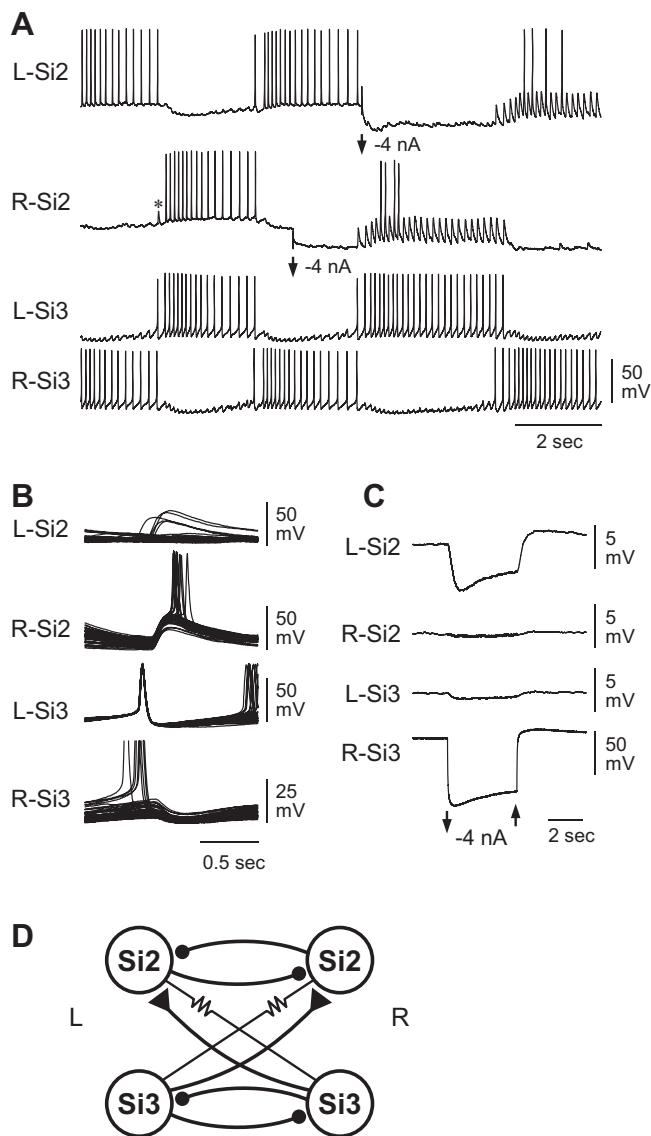


Fig. 8. Si3 drives Si2 via an excitatory synapse and an electrical connection. *A*: during the swim motor pattern, a hyperpolarization of Si2 by a negative current step (-4 nA) revealed excitatory synaptic drive from Si3. Each Si3 spike produced an EPSP in the contralateral Si2. Often, these EPSPs could be seen before the first spike of an Si2 burst (asterisk). *B*: superimposed traces triggered from L-Si3 spikes showed that the discrete EPSPs in R-Si2 and inhibitory postsynaptic potentials (IPSPs) in R-Si3 occurred at a constant latency, suggesting that they are monosynaptic. No time-locked synaptic potentials were observed in the L-Si2. *C*: a hyperpolarizing current pulse injected into Si3 produced a large hyperpolarization in the contralateral Si2. The ipsilateral Si2 and the contralateral Si3 also showed small changes in membrane potential, suggesting that all 4 neurons are electrically coupled. The recording was made in high-divalent cation (Hi-Di) saline to eliminate spontaneous spiking activity. *D*: a summary diagram of connections among Si2s and Si3s. In addition to the mutual inhibition between the Si2/Si3 pair, Si3 makes a strong excitatory synapse (triangle) and electrical connection (wavy line) onto the contralateral Si2.

Si3 was 0.009 ± 0.008 ($n = 8$). Thus the contralateral Si2 and Si3 are strongly coupled through both excitatory synapses and electrical connections (Fig. 8*D*).

To test whether these excitatory synaptic potentials were produced by electrical or chemical synaptic connections, Ca^{2+} -free saline was applied. The depolarizing potentials evoked by Si2 in the contralateral Si3 showed no change in Ca^{2+} -free

saline (Fig. 9*Aii*; $n = 7$), indicating that it is electrotonic in nature. In contrast, Ca^{2+} -free saline largely reduced the Si3-evoked excitatory synaptic potential in the contralateral Si2 (Fig. 9*B*). The remaining biphasic potential is most likely caused by the electrical coupling between Si3 and the contralateral Si2.

These results suggest that whereas the Si3 excitation of the contralateral Si2 was largely mediated by chemical synaptic transmission, the Si2 excitation of the contralateral Si3 was mediated by electrical transmission. The Ca^{2+} -free saline also revealed differences in the amplitudes of electrotonic potentials seen in Si2 and Si3. In Ca^{2+} -free saline, Si2-evoked depolarizing potentials in Si3 (Fig. 9*Aii*) were always larger than Si3-evoked electrotonic potentials in Si2 (Fig. 9*Bii*; Si2 to Si3, 1.3 ± 0.3 mV; Si3 to Si2, 0.23 ± 0.11 mV; $n = 7$; $P < 0.001$ by paired *t*-test). This conflicts with the observation that there was no rectification found at the electrical connection between contralateral Si2 and Si3 when measured with hyperpolarizing current injection (Fig. 8*C*). Since the EPSP-like depolarizing potentials were mediated by action potentials that propagated actively through the axon, the results suggest that the locus of the gap junction might be closer to the Si3 soma than the Si2 soma. Alternatively, it could indicate a greater degree of

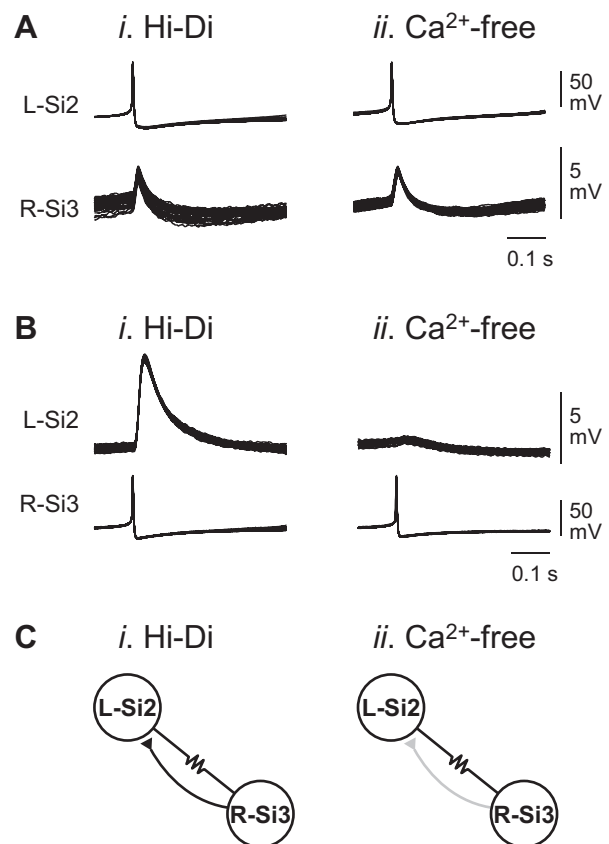
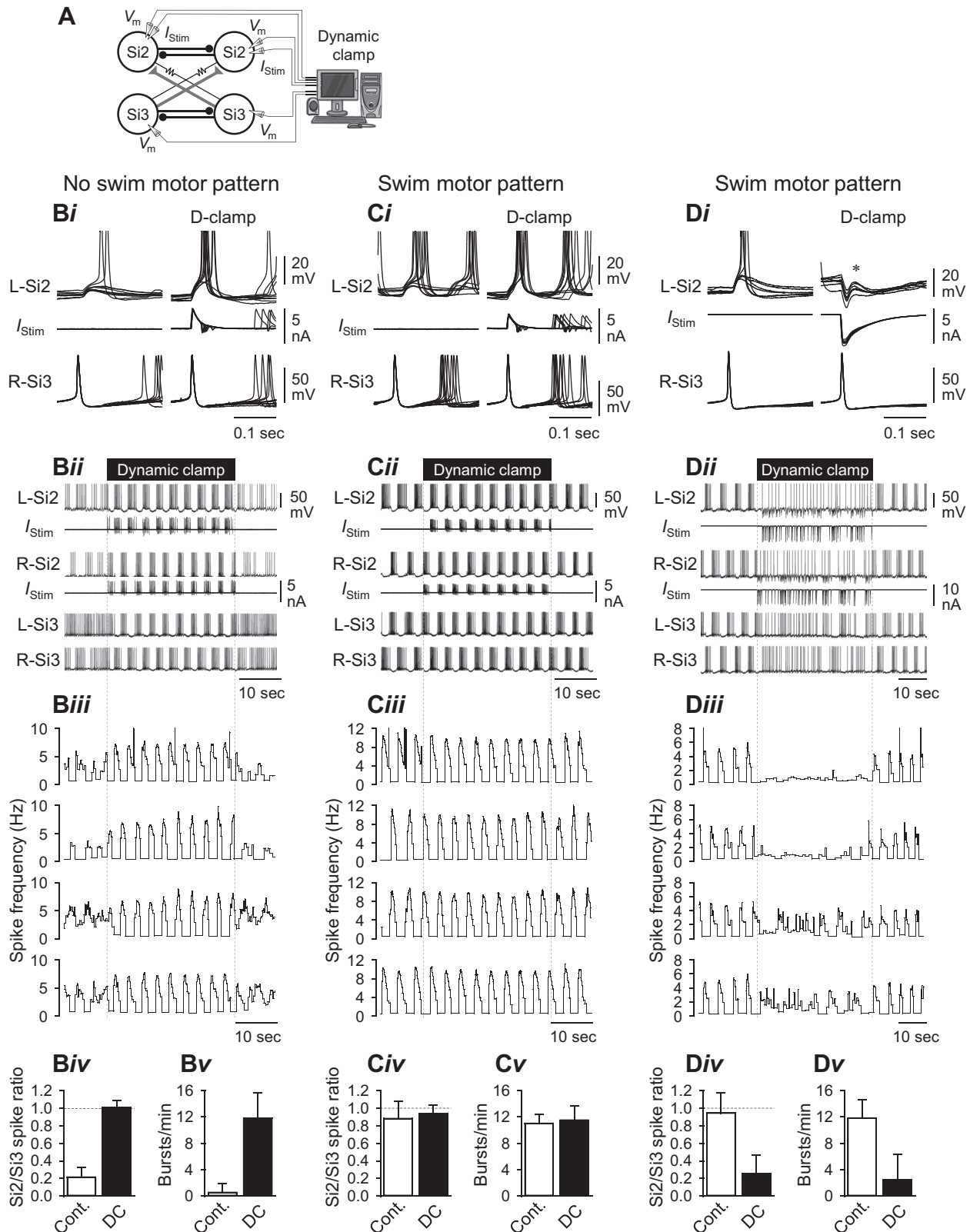


Fig. 9. The Si2-to-Si3 excitation is mediated by an electrical synapse, whereas Si3-to-Si2 excitation is mediated mainly by a chemical synapse. *Ai*: in high-divalent cation (Hi-Di) saline, spikes in L-Si2 evoked constant-latency EPSPs in R-Si3. *Aii*: in Ca^{2+} -free saline, the EPSP in R-Si3 was unchanged, suggesting that it was due to an electrical, not a chemical, synapse. *Bi*: in Hi-Di saline, overlaid action potentials in R-Si3 corresponded to constant-latency large EPSPs in L-Si2. *Bii*: in Ca^{2+} -free saline, the EPSP in L-Si2 was almost completely blocked. *Ci* and *Cii*: circuit diagrams summarizing the recording situation and results. Traces in *A* and *B* were obtained together from the same preparation.

low-pass filtering when the spike transits from Si3 to the contralateral Si2.

Si3-evoked synaptic drive of Si2 is necessary for motor pattern generation. To examine the role of the excitatory synapse from Si3 to Si2 in the generation of bursting activity,

we created an artificial synaptic conductance with the dynamic clamp technique, allowing us to perform “virtual pharmacology” experiments on the circuit (Fig. 10). During dynamic clamping, the amplitude and time course of the synaptic current was calculated in real time (see MATERIALS AND METH-



ODS) and was injected into the postsynaptic neuron through a second electrode (Fig. 10A).

When the Si3-evoked EPSPs were boosted by introducing an artificial synaptic conductance into Si2, it increased the probability of Si2 firing an action potential when Si3 fired (Fig. 10, *Bi* and *Biv*). In the experiment shown in Fig. 10B, a weak depolarizing current step (0.5 nA) was injected into both Si3s before dynamic clamping, which caused the Si3s to fire irregularly but was not sufficient to induce rhythmic bursting (Fig. 10*Bii*; cf. Fig. 7*Biii*). Under such conditions, boosting the Si3-to-Si2 synapse by introducing an artificial excitatory synapse induced rhythmic bursting in all four neurons (Fig. 10, *Bii* and *Biii*). The average Si2 burst frequency during dynamic clamp was 11.8 ± 3.8 bursts/min ($n = 6$; Fig. 10*Bv*), which was significantly larger than in the control (0.53 ± 1.3 bursts/min; $P = 0.001$ by paired *t*-test) but similar to the burst frequency during the swim motor pattern in the same preparation (11.1 ± 1.4 bursts/min; $P = 0.53$ by paired *t*-test).

During a nerve-evoked swim motor pattern, the probability of Si2 firing an action potential in response to a Si3 spike became much higher than when the preparation was at rest (0.9 ± 0.2 , $n = 7$; Fig. 10*Civ*). Under such conditions, boosting the synapse by dynamic clamp did not significantly increase the probability (0.94 ± 0.10 , $n = 7$; $P = 0.47$ by paired *t*-test; Fig. 10*Civ*) and did not further change the cycle frequency (Fig. 10*Cv*). The average Si2 burst frequency during dynamic clamp was 11.5 ± 2.2 bursts/min ($n = 7$; Fig. 10*Bv*), which was similar to the burst frequency during the swim motor pattern in the same preparation (11.0 ± 1.4 ; $P = 0.34$ by paired *t*-test).

The Si3-to-Si2 EPSPs could be counteracted by setting the artificial conductance to negative values, acting like a virtual antagonist. Although it did not fully cancel out the synaptic potentials, it successfully masked the excitatory synaptic drive from each Si3 to its contralateral Si2, preventing Si3 from causing Si2 to spike (Fig. 10*Di*). The probability of Si2 firing an action potential corresponding a Si3 spike during a swim motor pattern was significantly reduced from 0.94 ± 0.23 to

0.25 ± 0.21 ($P < 0.001$ by paired *t*-test; $n = 6$; Fig. 10*Div*). Counteracting the Si3-to-Si2 EPSP blocked the swim motor pattern; as soon as the dynamic clamp was turned on, Si2 and Si3 both stopped bursting and fired irregularly ($n = 4$ of 6; Fig. 10, *Dii* and *Diii*). During the dynamic clamping, Si3 continued firing in an irregular and slower left-right alternation. The average Si3 burst frequency during dynamic clamp was 2.4 ± 3.9 bursts/min ($n = 6$; Fig. 10*Dv*), which was significantly reduced from the burst frequency during the swim motor pattern in the same preparation (11.8 ± 2.7 bursts/min; $P = 0.007$ by paired *t*-test; $n = 6$; Fig. 10*Dv*). The swim motor pattern immediately resumed after the dynamic clamp was turned off. Together, these results show that the excitatory drive from Si3 to Si2 was critical for the generation of the swim motor pattern.

DISCUSSION

In this study, we described the network structure and function of a simple CPG consisting of only four neurons, which produces the alternating left-right motor output underlying the swimming behavior of the nudibranch, *D. iris*. In a previous study, we identified a bilaterally represented interneuron, Si2, that is a critical member of the swim CPG; the two Si2s form mutually inhibitory synapses and produce bursting activity in left-right alternation (Sakurai et al. 2011). In this study we discovered an additional bilaterally represented interneuron, Si3, which is also a member of this CPG, and demonstrated how the CPG functions.

Si2 and Si3 together form a half-center oscillator. Si3 has characteristics that allow it to be distinguished from Si2, including its soma size and location, synaptic connectivity, and effects on the swim motor pattern. Si3 is similar to Si2 in that it reciprocally inhibits its contralateral counterpart. The Si2 and Si3 pair form two layers of mutually inhibitory connections and act together as a half-center oscillator.

Reciprocal inhibition made by two neuronal elements, a so-called half-center oscillator, is one of the simplest forms of a network oscillator in which each neuronal element has no

Fig. 10. The use of “virtual pharmacology” at the Si3-to-Si2 synapse via dynamic clamp shows the importance of this synapse for rhythmic bursting. A: the experimental arrangement for dynamic clamp is shown (see MATERIALS AND METHODS for details). Every time the membrane potential (V_m) of Si3 surpassed a specified threshold (50% of spike height), an artificial synaptic current was calculated by the computer and injected (I_{stim}) into the contralateral Si2 to boost or negate the existing Si3-to-Si2 synapse, drawn in gray. In B–D, overlaid traces of membrane potential recordings from L-Si2 and R-Si3 and the stimulus current (I_{stim}) show the effect of dynamic clamp on the coupling of activity between Si2 and the contralateral Si3 (*i*). Simultaneous intracellular recordings from all 4 neurons (*ii*) and plots of their instantaneous spike frequency (*iii*) show the effect of dynamic clamping on their spiking activity. Bar graphs show the change in the probability of Si2 spike occurrence per Si3 spike (*iv*) and the burst frequency of R-Si2 (*v*) before (open bars) and during (filled bars) dynamic clamping. B: without an ongoing swim motor pattern, boosting the Si3-to-Si2 synapse with dynamic clamp initiated swimlike rhythmic bursting in all 4 neurons. In this condition, both Si3s were slightly depolarized by steady current (0.5 nA), producing only irregular spiking activity in the Si3s and corresponding EPSPs in Si2 before onset of the dynamic clamp. *Bi*: introducing a synaptic conductance (80 nS) with the dynamic clamp increased the EPSP amplitude (D-clamp; *right*). *Bii* and *Biii*: when the dynamic clamp was turned on, both Si2s and Si3s showed clear left-right alternation as in the swim motor pattern. When the dynamic clamp was terminated, bursting immediately stopped and the neurons started firing irregularly again. *Biv*: dynamic clamp caused a significant increase in the probability of Si2 firing following each Si3 spike ($P < 0.001$ by paired *t*-test, $n = 6$). *Bv*: dynamic clamp increased the burst frequency ($P < 0.001$ by paired *t*-test, $n = 6$). C: during an ongoing swim motor pattern, enhancement of the Si3-to-Si2 synapse had little effect. *Ci*: Si2 spikes became more time-locked to Si3 spikes when the dynamic clamp was on. *Cii* and *Ciii*: enhancing the Si3-to-Si2 synapse had little effect on the spiking or bursting activity of the ongoing motor pattern. *Civ*: the probability of Si2 spiking in response to each Si3 did not significantly increase the Si2/Si3 spike ratio which was already close to 1. *Cv*: enhancing the Si3-to-Si2 synapse had no significant effect on the burst frequency of ongoing swim motor patterns. D: during an ongoing swim motor pattern, negation of the excitatory synapse from Si3 to Si2 by dynamic clamp halted bursting. *Di*: when an artificial synaptic current with a negative conductance was introduced using dynamic clamp (*right*), it counteracted and reduced the amplitude of the actual synaptic potential, causing it to no longer recruit Si2 spikes (asterisk). *Dii* and *Diii*: as soon as the Si3-evoked EPSPs in Si2 were negated by dynamic clamp, rhythmic bursting halted in both Si2 and Si3. Rhythmic bursts in both neurons recovered immediately after cessation of dynamic clamp. *Div*: negation of the Si3-to-Si2 synapse significantly reduced the probability of Si2 firing in response to Si3 spikes. *Dv*: the burst frequency during dynamic clamp was significantly lower than cycle frequency of the swim motor pattern before dynamic clamp. In *Bi*, *Ci*, and *Di*, 10 consecutive sweeps were triggered by the peak of each Si3 spike and overlaid. Before the experiment, Si3 was depolarized to fire spikes by injecting a steady 1-nA current, and the synaptic conductance was increased up to 120 nS until every Si3 spike induced a spike in Si2 or decreased down to -240 nS until all Si2 spikes were blocked. Cont, control; DC, dynamic clamp.

endogenous rhythmicity (Brown 1911; Friesen 1994). Although it has been often described as a “building block” in CPGs (Getting 1989; Marder and Calabrese 1996), a number of studies in both vertebrates and invertebrates have shown that the half-center configuration plays the central role in rhythmic neural circuits for locomotion (Arshavsky et al. 1997, 1998; Friesen 1994; Friesen and Stent 1978; Getting 1989; Kiehn 2016; Marder and Calabrese 1996). Our results with current injection and dynamic clamp suggest that the four neurons of the CPG are nonbursting and that the two halves burst in alternation because of mutual inhibition. Thus the *Dendronotus* swim CPG is perhaps the simplest locomotor half-center oscillator described to date. Because of such simplicity, this circuit is also highly manipulable and hence may provide a good system to study the fundamental properties of a network oscillator.

An interesting aspect of the half-center organization of the *Dendronotus* swim CPG is that the components of each “half” of the oscillator are contralateral to each other. The contralateral Si3 and Si2 fire in phase with each other as a unit in part due to the strong synapse from Si3 to the contralateral Si2 and the electrical connection between them. The efferent motor activity recorded from the nerve root that would cause the animal to bend to the right was in phase with the right Si2 but was opposite to the right Si3. Conversely, the right Si3 and the left Si2 bursts are in phase with motor activity to the left. Thus, unlike the CPGs for vertebrate locomotion (Grillner 2006; Grillner and El Manira 2015; Kiehn 2016; Roberts et al. 1998), the functional organization of the half-center does not correspond to the anatomical locations of the interneurons.

Si2 and Si3 form a network oscillator. Neither Si2 nor Si3 showed endogenous oscillatory activity when activated alone. Simultaneous activation of all four neurons was necessary to generate rhythmic bursting that resembled the swim motor pattern. The bursting pattern started when current was injected, whereas it ceased immediately after the current injection was stopped. These results indicate that Si2 and Si3 have no endogenous bursting properties; they generate rhythmic bursts mainly through their synaptic interactions.

Originally, it was proposed that endogenous rhythmicity is not required in each neuronal element in the half-center configuration (Friesen 1994). However, few such circuits that have been described are true network oscillators. Instead, half-centers often exhibit endogenous bursting property when isolated (Arshavsky et al. 1986; Cymbalyuk et al. 2002; Selverston and Miller 1980). Conditional pacemaker properties also have been found in each half of the locomotor circuit in the spinal cord (Kiehn et al. 1996; Li et al. 2010; Reith and Sillar 1998; Wallen and Grillner 1987). It is possible that under some conditions, Si2 and Si3 may show bursting properties. Nonetheless, under our experimental conditions, there was no evidence for bursting properties. Thus the *Dendronotus* swim CPG may be a unique example of a true network oscillator in which each half-center has no endogenous rhythmicity.

Excitatory drive from Si3 to Si2 is necessary for rhythmic bursting. During an ongoing swim motor pattern, suppression of spiking in Si3 immediately halted the swim motor pattern. Artificial enhancement of the Si3 synapse using dynamic clamp initiated rhythmic bursting of both Si2 and Si3, but it had no effect on the ongoing swim motor pattern. Negation of the synaptic excitation of Si3 onto the contralateral Si2 by

dynamic clamping halted the bursting activity. These results indicate that synchronous excitation of Si2 together with the contralateral Si3 is necessary for the generation of rhythmic motor pattern.

In neural circuits with a half-center configuration as the main motif, each half of the neural network often contains excitatory components that provide rhythmic excitatory drive onto the mutually inhibitory neurons (Grillner 2006, 2003; Grillner and El Manira 2015; Hagglund et al. 2010, 2013; Hull et al. 2016; Kiehn 2016; Roberts et al. 1998, 2008b). In the mollusc *Clione*, rhythmic movement of winglike parapodia originates in two groups of neurons, 7 and 8, forming a half-center configuration in which one neuronal subtype makes excitatory synapses onto the other within group 8 (Arshavsky et al. 1985a). In vertebrates, ipsilaterally projecting excitatory interneurons have been suggested to play an important role in producing rhythmic drive within each half of the spinal cord (Christensen 1983; Grillner 2003; Kiehn 2016; Parker 2003; Roberts et al. 2008b). The importance of the ipsilateral excitatory neurons was also demonstrated optogenetically in the mammalian spinal cord, where activation of the excitatory neurons initiates locomotor-like activity, whereas inhibition of these neurons blocks it (Hagglund et al. 2010, 2013).

Neurons with both electrical connections and excitatory synapses often form a functional unit. For example, in the feeding network within the buccal ganglia of *Aplysia*, B31/2 and B63 are coupled together as a unit in the feeding circuit (Cropper et al. 2004; Hurwitz et al. 1997, 2003; Susswein et al. 2002). Mutual excitation and electrical coupling among the excitatory interneurons are necessary to synchronize activity among neurons and to sustain the antagonistic bursting activity produced by the reciprocally inhibitory components (Hinckley and Ziskind-Conhaim 2006; Li et al. 2009, 2010; Parker 2003; Parker and Grillner 1999; Perrins and Roberts 1995a, 1995b; Tresch and Kiehn 2000, 2002). In electric fish, electrotonic connections were seen among neurons that generate highly synchronized electric discharge in the medullary pacemaker nucleus (Moortgat et al. 2000a, 2000b). In mammals, it was suggested that electrical coupling among the excitatory Hb9 interneurons is important in coordinating the activity of this neuronal population in mature spinal networks (Hinckley and Ziskind-Conhaim 2006). In mammalian brain stem, electrical coupling and excitatory synapses in preBötzing complex are important for the generation of the breathing rhythm (Rekling et al. 2000).

Dynamic properties of CPG neurons and synapses. We have described the neuronal components and connections in the *Dendronotus* swim CPG. However, the generation of motor patterns is also dependent on dynamic properties that allow phase transitions; there must be a mechanism to either release the neurons from inhibition or to have them escape from inhibition (Arbas and Calabrese 1987a; Friesen 1994; Friesen and Stent 1978). Many studies have shown that neurons in a half-center oscillator often exhibit postinhibitory rebound (Arbas and Calabrese 1987a, 1987b; Arshavsky et al. 1985b, 1998; Dethier et al. 2015; Li and Moulton 2012; Perkel and Mulloney 1974; Quinlan and Murphy 1996; Satterlie 1985; Soffe et al. 2009; Syed et al. 1990). Studies using computer models also demonstrated the role of postinhibitory rebound (Daun et al. 2009; Perkel and Mulloney 1974; Wang and Rinzel 1992). In vertebrate locomotor networks, postinhibitory rebound was

suggested to play an important role in the maintenance of the rhythmic motor pattern (Li and Moulton 2012; Roberts et al. 2008a; Roberts and Tunstall 1990; Soffe et al. 2009).

In this study, we could not find any endogenous bursting properties in either Si2 or Si3. Moreover, although we demonstrated that all neurons need to be depolarized to generate strong bursting activity, it is still unclear how such tonic depolarization is provided to all of the CPG neurons during the normal production of the swim motor pattern. Further studies on the membrane and synapses may reveal the dynamic properties underlying their bursting activity.

Comparisons to the *Melibe* swim CPG. *Melibe leonina* is another nudibranch that also swims by rhythmically flexing its body from side to side (Watson et al. 2001). The CPG underlying the swimming behavior of *Melibe* has been worked out in detail (Sakurai et al. 2014; Thompson and Watson 2005). The *Melibe* swim CPG consists of four pairs of interneurons (Si1, Si2, Si3, and Si4). In contrast, in *Dendronotus* so far we have identified three pairs of interneurons (Si1, Si2, Si3) in this and the previous study (Sakurai et al. 2011). We previously suggested that Si1 in *Dendronotus* is homologous to Si1 in *Melibe*, based on their neuronal anatomy and immunohistochemical properties (Sakurai et al. 2011). In both species, Si1 resides near a cluster of serotonergic neurons in the cerebral ganglion, projects an axon ipsilaterally to the pedal ganglion, and exhibits FMRamide-like immunoreactivity.

In addition to the differences in the number of the swim interneurons found so far, there are substantial differences in the organization and functional activity of the two swim CPGs. First, the role of Si1 is different between the two species. Unlike in *Dendronotus*, in which Si1 serves as an extrinsic regulatory function, the *Melibe* Si1 acts as a member of the swim CPG, exhibiting bursting activity through mutual inhibition with the contralateral Si1 and Si2 and strong electrical coupling with the ipsilateral Si2 (Sakurai et al. 2011; Thompson and Watson 2005). Second, phase relationships among the CPG neurons differ between *Melibe* and *Dendronotus*. In *Melibe*, slow excitatory synaptic actions from both Si1 and Si2 induce a burst in Si3 with a 20% phase delay, which feeds back inhibitory synaptic actions to the Si1 and Si2 and contributes to the termination of their bursts (Sakurai et al. 2014). In contrast, in *Dendronotus* the contralateral Si2 and Si3 burst together via an excitatory synaptic connection from Si3 to Si2 and strong electrical coupling. Thus, unlike the *Melibe* swim CPG, which works with sequential actions of multiphasic recurrent synapses through more complex architecture with four pairs of neurons, including one in the cerebral ganglia, the *Dendronotus* swim CPGs consists of a simple half-center oscillator formed by Si2 and Si3.

With only Si1 and Si2 identified in the two species, we previously suggested that Si2 in *Dendronotus* was also homologous to Si2 in *Melibe*. Now that two additional pairs (Si3 and Si4) have been found in *Melibe* (Sakurai et al. 2014) and one additional pair (Si3) has been found in *Dendronotus* in the present study, it is less clear whether either Si2 or Si3 in *Dendronotus* is homologous to Si2, Si3, or Si4 in *Melibe*. However, the following characteristics may suggest that Si2 is homologous in the two species. First, the activity phases of Si2 are similar between these two species. In *Melibe*, it was shown that the burst phase of Si1 and Si2 correlates with the ipsilateral motor output (Lawrence and Watson 2002; Thompson and

Watson 2005; Watson et al. 2001, 2002). In *Dendronotus*, we observed that the Si2 burst was concurrent with the impulse burst in the ipsilateral PN3, whereas the Si3 bursts occurred in phase with the contralateral nerve activity. Thus the *Dendronotus* Si2 is functionally more similar to the *Melibe* Si2.

It is also possible that Si3 in *Dendronotus* is homologous to Si3 or Si4 in *Melibe* despite the functional and circuit differences. Further studies using gene expression may provide additional markers to allow us to more confidently determine neuronal homologies. Nonetheless, it is clear that although these two nudibranchs exhibit similar swimming behaviors and have some homologous neurons, the mechanisms underlying the generation of the motor pattern are substantially different. It is interesting how these two species have evolved such different neuronal configurations for such similar behaviors.

ACKNOWLEDGMENTS

We thank the anonymous referees for helpful comments.

GRANTS

This work was supported by National Science Foundation Grant IOS-1455527.

DISCLOSURES

No conflicts of interest, financial or otherwise, are declared by the authors.

AUTHOR CONTRIBUTIONS

A.S. and P.S.K. conception and design of research; A.S. performed experiments; A.S. and P.S.K. analyzed data; A.S. and P.S.K. interpreted results of experiments; A.S. and P.S.K. prepared figures; A.S. and P.S.K. drafted manuscript; A.S. and P.S.K. edited and revised manuscript; A.S. and P.S.K. approved final version of manuscript.

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